ONE ANNOTATED SLICE PER MOUSE: LABEL PROPAGATION FOR TRAINING AND ADAPTING BRAIN EXTRACTION NETWORKS IN PRECLINICAL MRI

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ABSTRACT

Brain extraction is the first step of most preclinical MRI pipelines, yet training a network for it, or adapting one to a new acquisition protocol, still requires every slice of every mouse to be outlined by hand. We ask how few annotated slices are enough when the remaining slices receive automatically propagated labels. On 20 mice from two acquisition protocols, we compare five propagators, including the video foundation model SAM2, under animal-level cross-validation. With one annotated slice per training mouse (9% of slices), a 2.5D network trained on SAM2-propagated labels reached a Dice of 0.950 and was non-inferior to full annotation (0.946; margin 0.01, $p < 0.001$), whereas training on that slice alone gave 0.891. nnU-Net behaved the same (0.958 against 0.960). A network trained on one protocol fell from 0.953 to 0.806 on the other in one direction; fine-tuning it with one annotated slice from each of two new mice plus SAM2 labels raised it to 0.928, matching fine-tuning on those mice fully annotated, while the two slices alone gave 0.866. Copying or interpolating masks was worse than no propagation, and agreement filtering helped registration but harmed SAM2.

***Index Terms*—** Brain extraction, mouse MRI, sparse annotation, label propagation, domain adaptation, SAM2

1. INTRODUCTION

Preclinical neuroimaging pipelines begin with brain extraction, which precedes atlas registration, volumetry and lesion assessment [1, 2]. Deep networks do this accurately, but training them requires masks drawn slice by slice, repeated whenever a lab changes scanner, protocol or disease model.

Self-supervised pretraining reduces this burden; masked image modelling improved mouse brain extraction in our previous work [3]. Sparse annotation labels only some slices [4] and fills in the rest, geometrically by shape interpolation [5] or slice-to-slice registration [6], or with a promptable video model such as SAM2 run through the slice stack [7, 8, 9]. SLM-SAM 2 propagates a few labelled slices through human CT and MR volumes [10], and nnSAM2 trains nnU-Net on SAM2 labels from one slice for paraspinal muscle [11]. For rodents, BEN adapts brain extraction to new scanners and species with a few fully labelled scans [12].

These studies evaluate SAM2-based propagation on human data, and adaptation has been measured in labelled scans rather than slices. It remains open whether propagated labels can replace full annotation in preclinical MRI, with thick slices and fast-changing brain shape, and how much annotation a new protocol costs. Our contributions are:

- A controlled comparison of five propagators, scoring the labels and the networks trained on them; copying and interpolation are worse than none.
- SAM2 labels from one annotated slice per mouse give a network non-inferior to full annotation, for two architectures.
- Adapting a trained network to a new protocol with one annotated slice per new mouse and SAM2 labels is non-inferior to fully annotating those mice; the slices alone recover about half as much.
- Agreement filtering helps a propagator near chance on disputed pixels and harms one that is mostly right.

2. METHODS

2.1. Data, annotation budget and network

We used 20 mice from a 9.4 T Bruker scanner (axial T1/T2-weighted, 1 mm slices), each with 10 or 11 slices (218 in total) and expert brain masks, in two cohorts of ten with matrices of 256×256 and 200×200. Volumes were percentile-clipped (0.5, 99.5), scaled to $[0, 1]$ and resampled to 256×256.

Only one to three slices of each training mouse, the anchors, were treated as annotated. Three anchors sat at $\lfloor n/4 \rfloor$, $\lfloor n/2 \rfloor$, $\lfloor 3n/4 \rfloor$ (or at 1, $\lfloor n/2 \rfloor$, $n-2$, which bounds the ends of the stack; 28% of slices), two at $\lfloor n/3 \rfloor$, $\lfloor 2n/3 \rfloor$ (18%) and one at $\lfloor n/2 \rfloor$ (9%). Validation and test mice kept full masks for model selection and scoring only.

All methods trained the same 2.5D CNN–Transformer (Fig. 1b; a slice and its two neighbours as input, a convolutional stem, a four-layer Transformer on 16×16 tokens and a decoder), for 8000 iterations with AdamW (3×10^{-4}, cosine decay, batch 8), a weighted cross-entropy plus Dice loss and standard augmentation, keeping the best validation checkpoint. Inference added a flipped pass and kept the largest

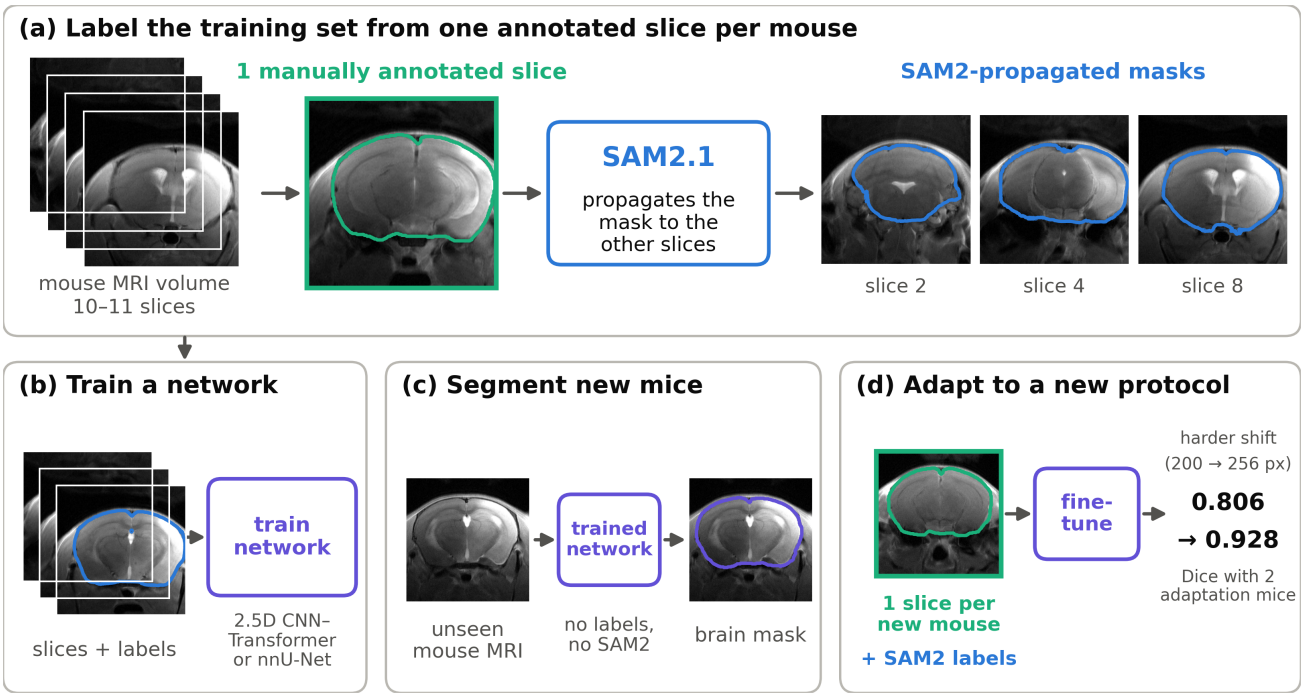


Fig. 1. Overview of the proposed framework. (a) SAM2 propagates labels from one annotated MRI slice per mouse. (b) Propagated labels train a brain extraction network. (c) The trained network segments unseen mice. (d) Sparse annotations enable adaptation to a new acquisition protocol.

component.

2.2. Propagators

SAM2. Each volume was treated as a video and SAM2.1 (Hiera-L) [7], without fine-tuning, was prompted with the anchor masks; forward and backward propagations were averaged.

Registration. A U-Net estimated displacement fields $\phi_{i\to i\pm1}$ between neighbouring slices at half resolution, trained with local normalised cross-correlation (window 9) and a smoothness penalty. An anchor-consistency loss requires the chained deformations to carry one anchor mask onto the next:

$$\mathcal{L}_{\text{anchor}} = 1 - \text{Dice}(M_a \circ \phi_{a\to a\pm1} \circ \cdots \circ \phi_{b\mp1\to b},\ M_b)\,. \tag{1}$$

Masks from the nearest anchors on both sides were averaged with inverse-distance weights. We also ran registration without $\mathcal{L}_{\text{anchor}}$.

Baselines. Copying the nearest anchor mask, shape-based interpolation of signed distance maps [5], and self-training [13] on the predictions of an anchor-only network. Non-SAM2 propagators used three anchors.

Pretraining and filtering. As an alternative to propagation, the encoder was pretrained by masked image modelling [3] on all training slices (4000 iterations) and fine-tuned on the anchors. For agreement filtering, a network trained on the anchors alone acted as a teacher, and pixels where it disagreed with the propagated label received zero loss weight. Registration kept the average of both maps; SAM2 kept its own label.

2.3. Adaptation to a new protocol

We trained the network with full annotation on eight mice of one cohort (two more for checkpoint selection) and fine-tuned it on $k \in \{1, 2, 4, 8\}$ mice of the other cohort, each with only its middle slice annotated. Target mice were split into five folds of two test mice, the first k of the other eight being used, so each was tested once per k. Fine-tuning ran for 2000 iterations at 10^{-4}, with half of each batch from source mice, and kept the final checkpoint. Target mice were supervised by the slice alone, with SAM2 labels, or with all slices annotated.

2.4. Evaluation

Cross-validation was five-fold at the animal level, stratified by cohort (14 training, two validation, four test mice per fold), with three seeds. Dice and slice-wise 95th-percentile Hausdorff distance (HD95, px) were averaged over seeds per mouse and compared with paired Wilcoxon tests. Non-inferiority used a one-sided paired Wilcoxon test with a 0.01

Table 1. Test results on 20 mice, three anchors per training mouse. Label: Dice of propagated labels on unannotated slices. *: different from anchors only ($p < 0.05$). †: not different from full annotation.

Method	Dice ↑	HD95 ↓	Label
Anchors only	0.924	13.2	–
+ copy nearest	0.908*	13.3	0.864
+ shape interpolation	0.909*	12.8	0.871
+ self-training	0.927*	12.6	0.893
+ registration	0.930	11.7	0.902
+ reg., anchor loss	0.931*	11.2	0.906
+ reg., anchor loss, agreement	0.933*	11.0	0.911
+ SAM2	**0.950***†	**8.9**	**0.940**
+ SAM2, agreement	0.940*	10.5	0.940
All slices annotated	0.946	9.0	–

Table 2. Annotation budget: test Dice and HD95 (px) on 20 mice with one to three annotated slices per training mouse, alone or with SAM2 labels. Label: Dice of SAM2 labels. †: non-inferior to full annotation ($p < 0.001$). n/a: not applicable. MIM: masked image pretraining, one slice (separate runs; own control 0.892 / 20.6).

		Annotated only		+ SAM2		
Slices	%	Dice	HD95	Dice	HD95	Label
1	9	0.891	20.4	**0.950**†	9.0	0.942
2	18	0.925	13.4	0.949†	8.9	0.949
3	28	0.924	13.2	0.950†	8.9	0.940
All	100	0.946	9.0	n/a	n/a	n/a
1 + MIM	9	0.890	21.4	n/a	n/a	n/a

Dice margin, 2.4 times the seed-to-seed standard deviation (0.004). nnU-Net v2 [14] (2D, 250 epochs, final checkpoint, same post-processing) was trained once per fold with full annotation, one slice plus SAM2 labels, or one slice with the rest ignored.

3. RESULTS

One slice with SAM2 matches full annotation. With three anchors (Table 1), the anchor-only network reached 0.924 against 0.946 with every slice annotated ($p < 0.001$). SAM2 labels raised it to 0.950 and 8.9 px, better than anchors only and than the best registration variant on 19 of 20 mice each ($p < 0.001$), and not different from full annotation in Dice ($p = 0.96$) or HD95 ($p = 0.76$), in both cohorts. With one annotated slice (Table 2), training on it alone gave 0.891 and 20.4 px, while SAM2 labels gave 0.950 and 9.0 px, better on all 20 mice and non-inferior to full annotation ($p < 0.001$). Masked image pretraining did not help (0.890 with one slice; 0.922 against 0.925 with three).

Naive propagation hurts. Copying and interpolation lowered Dice to 0.908 and 0.909, below anchors only. Their labels fell from 0.92–0.93 next to an anchor to 0.705 on the end slices beyond the last anchor (Fig. 2), against 0.726 for self-training, 0.786 for registration and 0.883 for SAM2. Anchors at the stack ends helped interpolation, self-training and filtered registration ($p < 0.01$), but not copying, unfiltered registration or SAM2.

Table 3. Adaptation: a network trained on one cohort is fine-tuned on k mice of the other and tested on its held-out mice (10 mice, three seeds). Target labels: the middle slice of each mouse, plus SAM2 labels, or all slices. †: non-inferior to all slices at a 0.01 Dice margin ($p < 0.05$). Rows for $k = 1$ come from separate runs, whose unadapted networks gave 0.930 and 0.804.

		256→200		200→256	
k	Target labels	Dice	HD95	Dice	HD95
0	none	0.932	11.3	0.806	33.1
1	slice	0.932	11.3	0.859	25.2
	+ SAM2	0.937†	11.0	0.910†	17.8
	all	0.930	10.5	0.910	17.6
2	slice	0.934	11.0	0.866	24.1
	+ SAM2	0.941†	9.7	0.928†	14.6
	all	0.941	9.1	0.928	14.6
4	slice	0.935	10.9	0.873	22.5
	+ SAM2	0.943†	9.4	0.941†	11.8
	all	0.940	9.1	0.944	11.3
8	slice	0.936	10.6	0.882	21.4
	+ SAM2	**0.945**†	**8.8**	0.948†	10.2
	all	0.941	9.0	**0.951**	**9.5**

Filtering depends on the propagator. Filtering ignored 2–3% of unannotated pixels. It gave the best registration variant but reduced SAM2 to 0.940 ($p < 0.001$): registration had labelled 57–60% of the ignored pixels correctly, near chance, but SAM2 72–78%, so filtering removed correct boundary supervision.

nnU-Net. With full annotation nnU-Net reached 0.960 and 6.9 px, 0.013 above our network. From one slice it fell to 0.868 alone but reached 0.958 and 7.6 px with SAM2 labels, non-inferior to full annotation ($p < 0.001$).

One slice per new mouse adapts to a new protocol. Trained on one cohort, the network lost little on the 200×200 cohort (0.932) but fell to 0.806 and 33.1 px on the 256×256 cohort, against 0.953 within it (Table 3). One annotated slice from one target mouse plus SAM2 already gave 0.910, equal to annotating that mouse fully and above the slice alone (0.859, $p = 0.002$). With two mice, the slices alone gave 0.866 and with SAM2 labels 0.928, better on all ten test mice ($p = 0.002$) and non-inferior to annotating both mice fully (0.928, $p = 0.003$). Four and eight mice gave 0.941 and 0.948 against 0.944 and 0.951, both non-inferior; slices alone

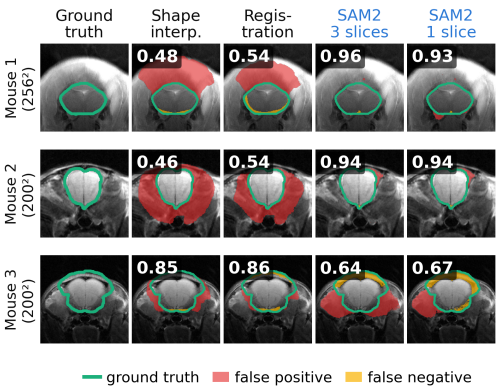


Fig. 2. Qualitative comparison of label propagation methods on an end slice of three training mice. Rows 1–2 show successful SAM2 propagation and row 3 a failure case. Columns 2–4 use three annotated slices (interp.: shape interpolation), the last column one. Green: ground truth; red: false positive; orange: false negative. Numbers: Dice of each propagated label against the ground truth.

reached only 0.882 with eight. In the milder direction SAM2 labels were non-inferior to full annotation for every k and beat the slices alone for $k \geq 2$ ($p \leq 0.010$).

4. DISCUSSION AND CONCLUSION

One annotated slice per mouse and a promptable video model replaced slice-by-slice annotation for training a brain extraction network, with 91% fewer annotated slices and no loss beyond a 0.01 Dice margin. The same recipe adapted a network to a new protocol from one annotated slice per new mouse, as well as full annotation of those mice. A propagator that cannot follow anatomy beyond the annotated slices is worse than none, and agreement filtering helps only a propagator near chance where it acts.

Limits: 20 mice from one scanner, whole-brain masks only, and annotation time was not measured. Cross-validation used two fully annotated validation mice per fold. Adaptation was tested between two protocols of one scanner and should be confirmed across sites.

5. COMPLIANCE WITH ETHICAL STANDARDS

[TODO: Animal study approval: institution and protocol number (IACUC) for the MRI data, from coauthors.]

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